SUSTAINABILITY



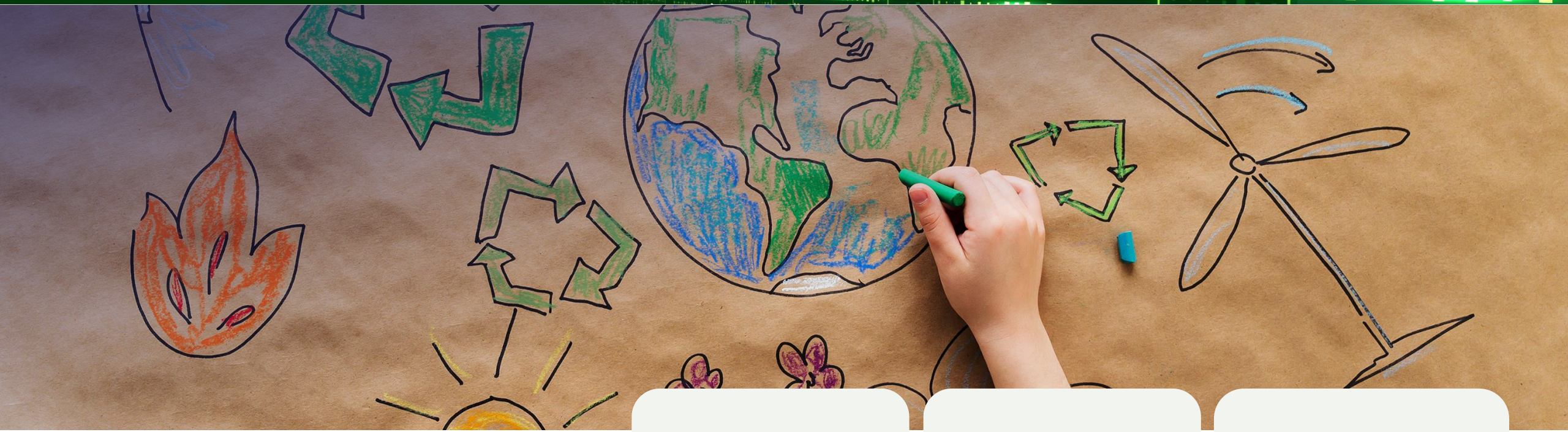
Enabling Scalable Waste Heat Recovery

Operational Realities, Design Principles, and Value from the Datacenter Perspective

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Datacenter waste heat recovery (WHR) can create *operational*, *economic*, and *societal* value.

Problem

Misaligned expectations prevent WHR from scaling.

Solution

Align stakeholders, optimize the system, and balance costs and benefits.

Result

Repeatable systems that enable broader deployment and improve energy-system efficiency.

A Common Framework for Waste Heat Recovery

Scalable WHR requires alignment across heat supply, recovery, delivery, and demand.



1. Generate

Datacenter operations produce heat as a byproduct of computation.



2. Recover

Heat is captured and transferred at a defined recovery interface.



3. Deliver

Thermal energy is transported to an offtaker.



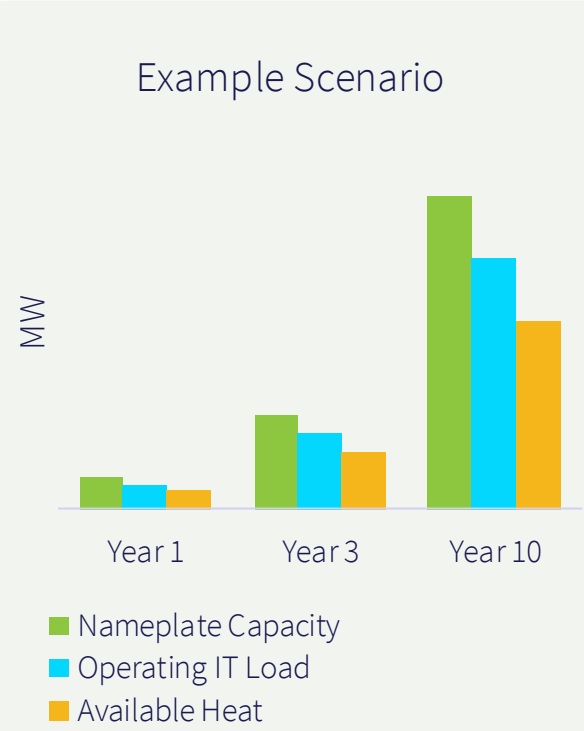
4. Utilize

The offtaker uses the heat to meet a thermal demand.

Available Heat Follows Operating IT Load

Available waste heat is determined by operating IT load, not nameplate campus capacity.

- **Nearly all IT power becomes heat.** As a rule of thumb, 1 MW of IT load produces approximately 1 MW of waste heat.
- **Nameplate capacity is the ceiling.** Actual heat generation depends on deployed capacity and utilization.
- **Operating IT load ramps over time.** Heat availability increases as halls are energized, equipment is deployed, and capacity is utilized.
- **Not all generated heat is practically recoverable.** Recovery must maintain required cooling-system operating conditions and datacenter reliability.

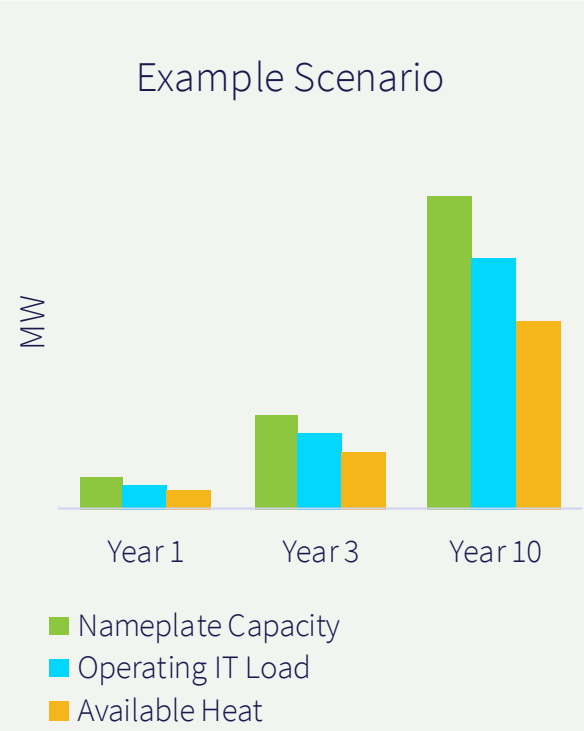


Takeaway: *Phase and size WHR infrastructure around dependable heat availability to avoid overbuilding.*

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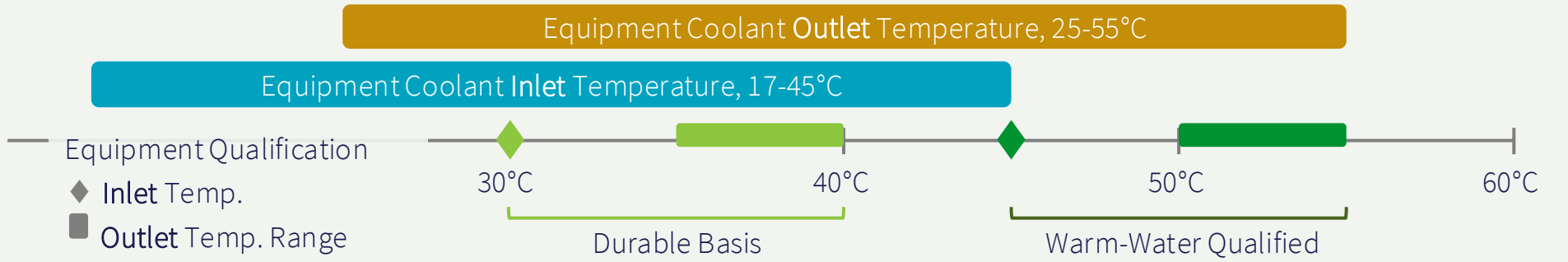


Takeaway: *Phase and size WHR infrastructure around dependable heat availability to avoid overbuilding.*

Cooling Design and Recovery Location Set Temperature

IT equipment temperatures influence, but do not equal, temperature available for recovery.

- **Equipment requirements set starting temperature.** Coolant must enter within the qualified range of the IT equipment.
- **IT equipment raises the coolant temperature.** Coolant ΔT determines the temperature leaving the equipment.
- **The cooling path determines what reaches the recovery interface.** Cooling architecture, heat exchangers, and recovery location affect the available temperature.

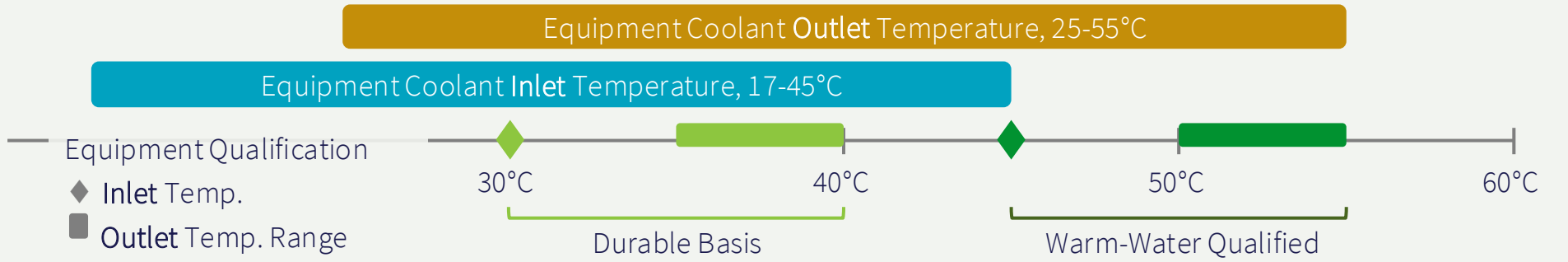


Takeaway: *Plan for <25–55°C heat source, then validate the site-specific temperature.*

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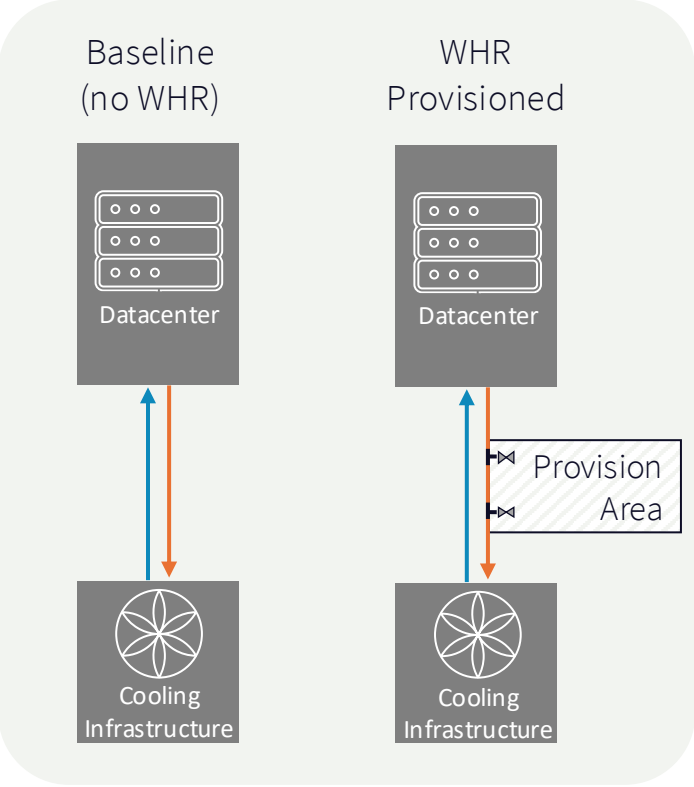


Takeaway: *Plan for <25–55°C heat source, then validate the site-specific temperature.*

Pre-Chiller Recovery Balances System Objectives

The best recovery point balances reliability, efficiency, and offtaker value, not temperature alone.

- **Minimize operational risk.** Pre-chiller recovery avoids direct integration with IT equipment and preserves the primary cooling path.
- **Create a cooling-load offset.** Heat supplied to the offtaker reduces the heat the datacenter needs to reject.
- **Provide consistent low-grade heat.** The recovery point provides the offtaker with a defined, usable heat source while preserving datacenter control.
- **Enable phased deployment.** The recovery connection can be pre-positioned during construction and activated as viable demand develops.

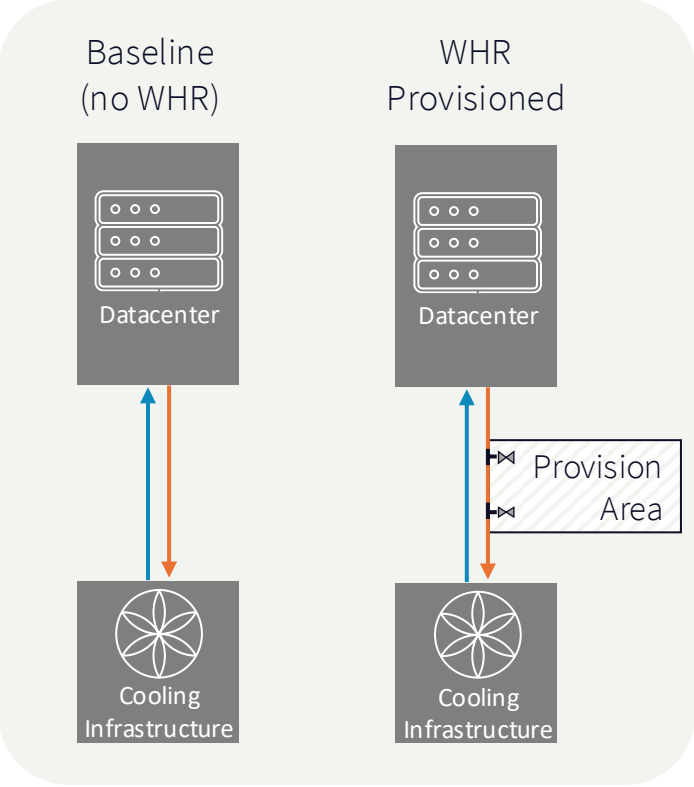


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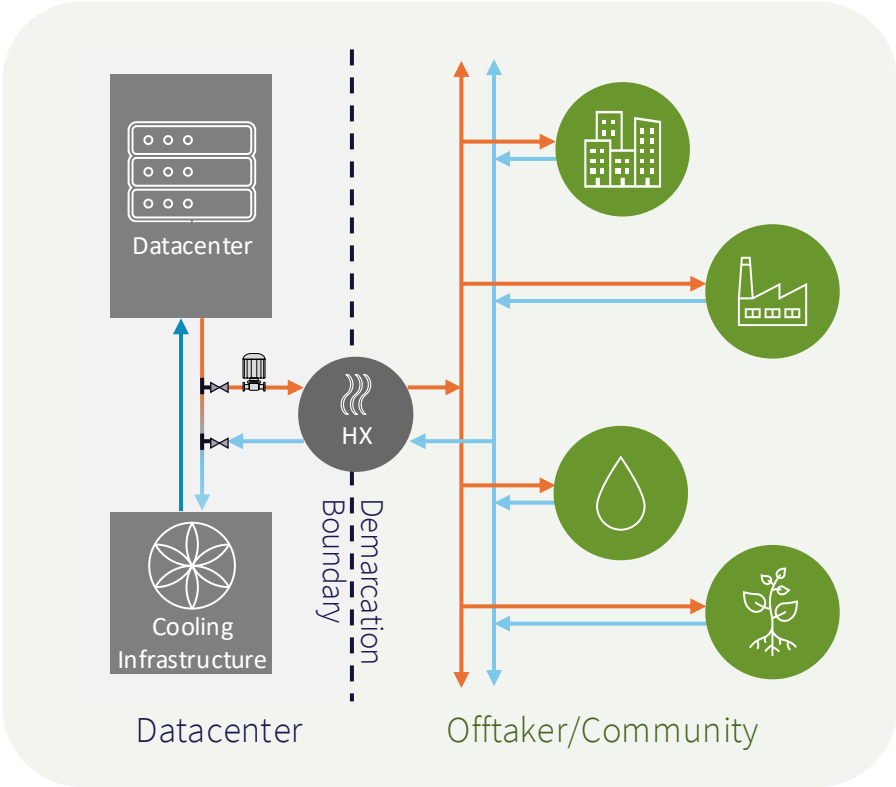


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The Heat Exchanger Defines the Demarcation Boundary

The heat exchanger transfers heat while separating datacenter and offtaker systems.

- **Deliver heat to the boundary.** Pumps circulate recovered heat from the recovery point to the heat exchanger.
- **Protect operations.** The heat exchanger hydraulically isolates the systems, limiting the impact of disruptions on datacenter cooling.
- **Clarify responsibilities.** The datacenter is responsible for infrastructure to the heat exchanger; the offtaker is responsible for downstream equipment, including any required heat pump.
- **Standardize delivery.** A consistent interface can support different sites and offtakers.



Takeaway: *A standardized boundary enables collaboration while preserving operational independence.*

Viability Depends on Matching Demand to Supply

The economics depend on how effectively the offtaker can use the available heat.

- **Match temperature.** Using heat at the supplied temperature avoids costly temperature upgrades and additional energy consumption.
- **Match timing.** Year-round demand increases utilization of the pipes, heat exchangers, and other installed infrastructure.
- **Match scale.** Meaningful heat use increases cooling offset and delivered energy while improving economies of scale.
- **Match location.** Nearby demand reduces distribution infrastructure, heat losses, and delivery costs.



Takeaway: *Match demand to supply to maximize value and avoid underutilized infrastructure.*

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Design Principles for Scalable Systems

1

Protect Core Operations

Preserve datacenter reliability and operational independence.

2

Create Shared Value

Balance costs, benefits, risks, and responsibilities across stakeholders.

3

Design for Repeatability

Standardize interfaces while adapting to site conditions.

Aligned Systems Can Create Shared Value



Datacenters

- Improve operational efficiency
- Address heat-reuse requirements



Offtakers

- Access a local thermal-energy resource
- Potentially lower energy costs and emissions



Communities

- Reduce demand for primary energy
- Improve overall energy-system efficiency

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Help Build a Common Framework for Scalable WHR

Problem to Solve

Misaligned assumptions and bespoke approaches limit viable partnerships and repeatable deployment.

Get Involved

- Join the OCP Heat Reuse Subproject.
- Contribute operational experience, offtaker requirements, design guidance, economic analysis, and project lessons.

Contribution Timeline

White paper on Enabling Scalable Waste Heat Recovery to be submitted for community review and feedback following the 2026 OCP Global Summit.

Learn More

OCP Heat Reuse Subproject: https://www.opencompute.org/community/heat-reuse